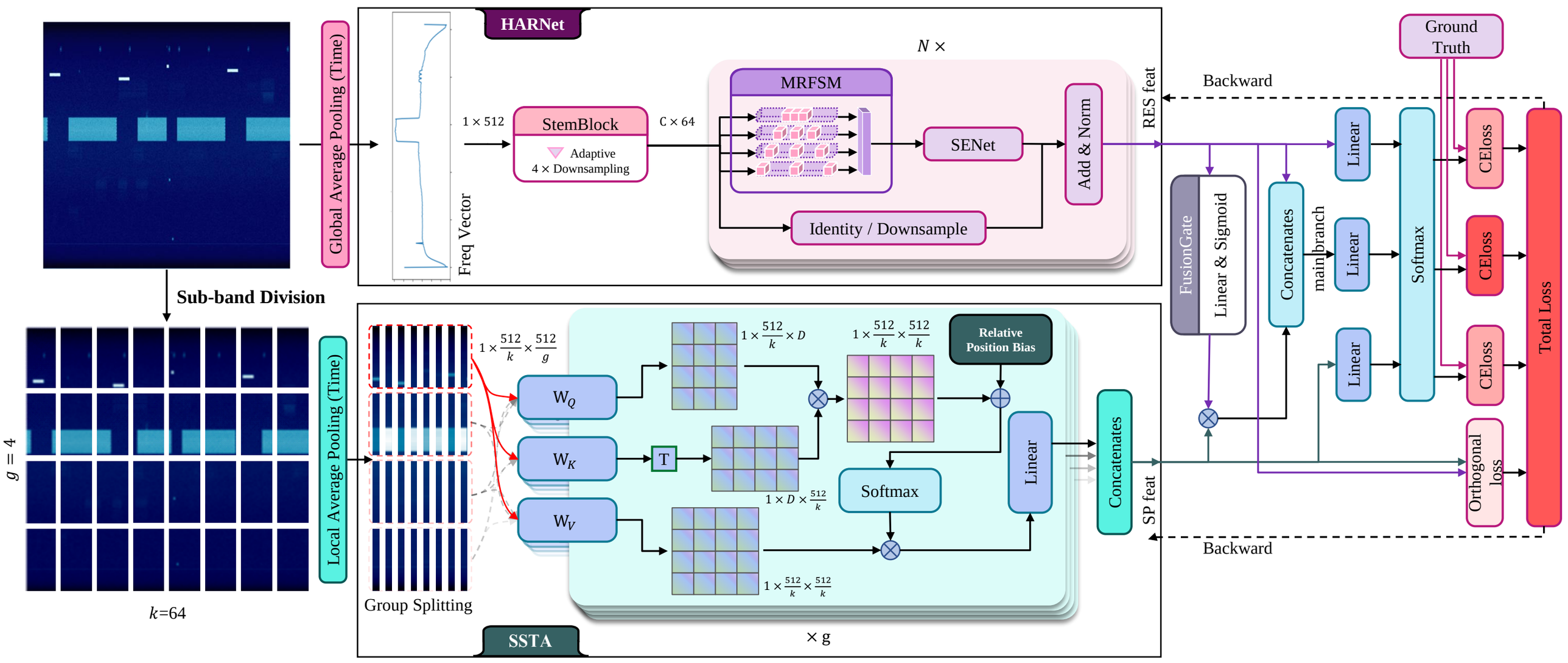

HARNet
N ×
Global Average Pooling (Time)
Freq Vector
1 × 512
StemBlock
Adaptive
4 × Downsampling
C × 64
MRFSM
SENet
Add & Norm
Identity / Downsample
RES feat
Backward
Ground Truth
Linear
Linear
Linear
Softmax
CEloss
CEloss
CEloss
Orthogonal loss
Total Loss
FusionGate
Linear & Sigmoid
Concatenates
main branch
Sub-band Division
g = 4
k=64
Local Average Pooling (Time)
Group Splitting
W_Q
W_K
W_V
T
Softmax
Relative Position Bias
Linear
Concatenates
SP feat
Backward
SSTA
× g